

A .
(Square $n \times n$)

$$\underline{A} \underline{x} = \lambda \underline{x} \quad (\text{non-zero } \underline{x}, \text{ numbers } \lambda).$$

$$\underline{A} \underline{x} - \lambda \underline{I}_n \underline{x} = \underline{0}$$
$$(\underline{A} - \lambda \underline{I}_n) \underline{x} = \underline{0} \quad \checkmark$$

A has free cols iff $\det(A) = 0$.

If A has free cols; say $\underline{c}_j = \sum_{i < j} \alpha_i \underline{c}_i$

$$\det \begin{bmatrix} | & & | & | & | \\ \underline{c}_1 & \dots & \underline{c}_{j-1} & \underline{c}_j & \dots & \underline{c}_n \\ | & & | & | & | \end{bmatrix} = \det \begin{bmatrix} | & & | & & | \\ \underline{c}_1 & \dots & \underline{c}_{j-1} & \alpha_1 \underline{c}_1 + \alpha_2 \underline{c}_2 + \dots + \alpha_{j-1} \underline{c}_{j-1} & \underline{c}_{j+1} \dots \underline{c}_n \\ | & & | & & | \end{bmatrix}$$
$$= \sum_{i < j} \alpha_i \det \begin{bmatrix} | & & | & | & | \\ \underline{c}_1 & \dots & \underline{c}_{j-1} & \underline{c}_i & \underline{c}_{j+1} \dots \underline{c}_n \\ | & & | & | & | \end{bmatrix} = 0.$$

If $\det(A) = 0$ then it has free cols.

OR If A has no free cols, then $\det(A) \neq 0$.

$\det(A) = \pm$ product of pivots $\neq 0$.

depending on how many row exchanges in elimination.

Check $\det(A - \lambda I) = 0$.

$$A = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}$$

$$A - \lambda I = \begin{pmatrix} 1-\lambda & 0 \\ -1 & 1-\lambda \end{pmatrix}$$

$$\lambda I = \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}$$

$\det(A - \lambda I) = ?$ use linearity properties.

First row $(1-\lambda) \begin{pmatrix} 1 & 0 \end{pmatrix} + 0 \begin{pmatrix} 0 & 1 \end{pmatrix}$

$$\det(A - \lambda I) = \det \begin{pmatrix} 1-\lambda & [1 \ 0] \\ -1 & (1-\lambda) \end{pmatrix} + \det \begin{pmatrix} 0 & [0 \ 1] \\ -1 & 1-\lambda \end{pmatrix}$$

$$= (1-\lambda) \det \begin{pmatrix} 1 & 0 \\ -1 & 1-\lambda \end{pmatrix} + 0 \cancel{\det \begin{pmatrix} 0 & 1 \\ -1 & 1-\lambda \end{pmatrix}}$$

Second row: $-1 [1 \ 0] + (1-\lambda)[0 \ 1]$

$$(1-\lambda) \det \begin{bmatrix} 1 & 0 \\ -1 [1 \ 0] + (1-\lambda)[0 \ 1] \end{bmatrix} = (1-\lambda) \det \begin{bmatrix} 1 & 0 \\ -1 [1 \ 0] \end{bmatrix} + (1-\lambda) \det \begin{bmatrix} 1 & 0 \\ (1-\lambda)[0 \ 1] \end{bmatrix}$$

$$= (1-\lambda)(-1) \det \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} + (1-\lambda)^2 \det \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \underline{(1-\lambda)^2}$$

$$A = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} \quad \det(A - \lambda I) = \underline{(1-\lambda)^2}$$

roots of $\det(A - \lambda I) = 0$ are 1, 1.

$$A - 1I = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} - 1 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ -1 & 0 \end{bmatrix}$$

$$\text{null}(A - I) = \left\{ \alpha \begin{pmatrix} 0 \\ 1 \end{pmatrix} : \alpha \in \mathbb{R} \right\}$$

$$\text{rref}(A - I) = \text{rref} \begin{bmatrix} 0 & 0 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\underline{C2} = \underline{0C1}$$

$$\underline{C2} - \underline{0C1} = \underline{0}$$

characteristic polynomial.

Roots of $\det(A - \lambda I) = 0$: Eigenvalues.

If a root is repeated k times in $\det(A - \lambda I) = 0$, we say that eigenvalue has algebraic multiplicity k .

Square Symmetric real A :

$$A^T = A \quad \text{eg: } A = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}.$$

$$A - \lambda I = \begin{pmatrix} 1-\lambda & -1 \\ -1 & 1-\lambda \end{pmatrix}.$$

$$\det(A - \lambda I) = (1-\lambda)^2 - 1 = \lambda^2 - 2\lambda + 1 - 1 = \lambda^2 - 2\lambda.$$

Roots: 0, 2.

$$\text{null}(A - 0I) = \text{null}(A) = \left\{ \alpha \begin{pmatrix} 1 \\ 1 \end{pmatrix} : \alpha \in \mathbb{R} \right\}.$$

$$\text{rref}(A) = \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix}, \quad \begin{array}{l} \underline{c2} = -\underline{c1} \\ \underline{c2} + \underline{c1} = 0. \end{array}$$

$$\text{null}(A - 2I) = \text{null}\left[\begin{pmatrix} -1 & -1 \\ -1 & -1 \end{pmatrix}\right] = \left\{ \alpha \begin{pmatrix} 1 \\ -1 \end{pmatrix} : \alpha \in \mathbb{R} \right\}.$$

Dot product of $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ -1 \end{pmatrix} = 0$.

The eigenvectors (one from each of $\text{null}(A)$ and $\text{null}(A - 2I)$)
 $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$ form a max linearly indep set in $\mathbb{R}^2$.

$$\text{For all } \underline{x} \in \mathbb{R}^2, \underline{x} = \alpha \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \beta \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

$$A \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 0 \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$A \begin{pmatrix} 1 \\ -1 \end{pmatrix} = 2 \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

$$A \underline{x} = A \left[\alpha \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \beta \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right].$$

$$= A \left[\alpha \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right] + A \left[\beta \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right].$$

$$= \alpha A \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \beta A \begin{pmatrix} 1 \\ -1 \end{pmatrix} =$$

$$= \alpha \cdot 0 \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix} + 2\beta \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

$$A(A \underline{x}) = 4\beta \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

$$B = \begin{pmatrix} 1 \\ -1 \end{pmatrix} \begin{pmatrix} 1 & -1 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} + \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix}$$

$$= \begin{pmatrix} 3/2 & 1/2 \\ 1/2 & 3/2 \end{pmatrix}.$$

$$\lambda I = \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}$$

Eigenvalues of B.

$$\det(B - \lambda I) = \det \begin{pmatrix} 3/2 - \lambda & 1/2 \\ 1/2 & 3/2 - \lambda \end{pmatrix}.$$

$$\begin{bmatrix} \frac{3}{2} - \lambda & 1/2 \\ 1/2 & 3/2 - \lambda \end{bmatrix} \xrightarrow{r_2 \leftarrow r_2 - \frac{1}{2} r_1} \begin{bmatrix} 3/2 - \lambda & 1/2 \\ 0 & \left(\frac{3}{2} - \lambda\right) - \frac{1}{2} \cdot \frac{1}{2} \end{bmatrix} \quad \lambda \neq 3/2.$$

$$\det(B) = \left(\frac{3}{2} - \lambda\right) \left(\frac{3}{2} - \lambda\right) - 1/4 = \lambda^2 - 3\lambda + 9/4 - 1/4 = \lambda^2 - 3\lambda + 2 = (\lambda - 2)(\lambda - 1).$$

Roots: 2, 1

$$\lambda = 2 \quad B - \lambda I = \begin{pmatrix} \frac{3}{2} - \lambda & 1/2 \\ 1/2 & 3/2 - \lambda \end{pmatrix} = \begin{pmatrix} -1/2 & 1/2 \\ 1/2 & -1/2 \end{pmatrix} \quad \begin{array}{l} \underline{c2} = -\underline{c1} \\ \underline{c1} + \underline{c2} = 0 \end{array}$$

$\therefore$ Eigenvector for eigenvalue $\lambda = 2$ $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

$$\lambda = 1 \quad B - \lambda I = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix} \quad \underline{c2} = \underline{c1}.$$

Eigenvector for eigenvalue $\lambda = 1$ $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$.

$$\underline{x} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

$$B \underline{x} = \frac{1}{2} B \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \frac{1}{2} B \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \frac{1}{2} \cdot 2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \frac{1}{2} \cdot 1 \cdot \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix} \text{ is in null } (B - 2I) \quad (B - 2I) \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 0. \quad B \begin{pmatrix} 1 \\ 1 \end{pmatrix} - 2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 0.$$

$$B \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 2 \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$B \begin{pmatrix} 1 \\ -1 \end{pmatrix} = 1 \cdot \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$B(Bx) = B\left[\frac{1}{2} \cdot 2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \frac{1}{2} \cdot 1 \begin{pmatrix} 1 \\ -1 \end{pmatrix}\right] = \frac{1}{2} \cdot 2^2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \frac{1}{2} \cdot 1^2 \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

$$B^n(x) = \frac{1}{2} \cdot 2^n \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \frac{1}{2} \cdot (1)^n \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Spectral Decomposition Theorem.

Suppose A is square, symmetric. $n \times n$ matrix.

(i) A has n real eigenvalues (possibly repeated).

$\det(A - \lambda I)$ has n real roots (possibly repeated).

(ii) Eigenvectors corresponding to distinct eigenvalues are orthogonal.

(iii) Suppose λ^* is an eigenvalue $(A - \lambda^* I)$ has free columns.

$$\dim[\text{null}(A - \lambda^* I)] = \# \text{ free cols in } A - \lambda^* I$$

$$= \# \lambda^* \text{ is repeated among roots of } \det(A - \lambda I)$$

$\Rightarrow$ we get n linearly indpt eigenvectors.

$\therefore$ eigenvectors form a basis for $\mathbb{R}^n$.

we can also get n orthogonal vectors as a basis for $\mathbb{R}^n$.

Normalize all so that they have length 1.

$$\underline{v}_1, \dots, \underline{v}_n.$$

$$A \underline{v}_1 = \lambda_1 \underline{v}_1,$$

$$A \underline{v}_2 = \lambda_2 \underline{v}_2$$

$$\vdots$$

$$A \underline{v}_n = \lambda_n \underline{v}_n.$$

$$\underline{v}_i^T \underline{v}_j = \begin{cases} 0 & \text{if } i \neq j \\ 1 & \text{if } i = j. \end{cases}$$

$$V = \begin{bmatrix} | & & | \\ \underline{v}_1 & \dots & \underline{v}_n \\ | & & | \end{bmatrix}.$$

$$V^T V = \begin{bmatrix} i,j^{\text{th}} \text{ entry} \\ = \underline{v}_i^T \underline{v}_j \end{bmatrix}$$

$$\underline{V}^{-1} = \underline{V}^T.$$

$$= \begin{bmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{bmatrix}$$

$$AV = A \begin{bmatrix} v_1 & \dots & v_n \end{bmatrix} = \begin{bmatrix} \lambda_1 v_1 & \lambda_2 v_2 & \dots & \lambda_n v_n \end{bmatrix}$$

$$= V \underbrace{\begin{bmatrix} \lambda_1 & 0 & & 0 \\ 0 & \lambda_2 & & \\ \vdots & \vdots & \ddots & \\ 0 & 0 & & \lambda_n \end{bmatrix}}_{\Lambda}$$

$$= V \Lambda$$

$$A \underbrace{V V^T}_{I} = V \Lambda V^T$$

$$\therefore A = V \Lambda V^T$$

$$\det(A - \lambda I) = 0.$$

Roots

$$\lambda_1 \geq \lambda_2 \geq \lambda_3 \dots \geq \lambda_n.$$

Distinct roots

$$\left\{ \begin{array}{l} \lambda_{i_1} \text{ repeats } n_1 \text{ times.} \\ \lambda_{i_2} \text{ repeats } n_2 \text{ times.} \\ \vdots \\ \lambda_{i_k} \text{ ——— } n_k \text{ times.} \end{array} \right.$$

$$i_1=1 \quad i_2=2 \quad i_3=4 \quad i_4=5$$

$$\lambda_1 \geq \lambda_2 = \lambda_3 > \lambda_4 > \lambda_5 = \lambda_6$$

$$1, 0.5, 0.5, 0.2, -0.5, -0.5$$

$$\lambda_1 \text{ has multiplicity } 1.$$

$$\lambda_2 \text{ has ——— } 2.$$

$$\lambda_4 \text{ has ——— } 1.$$

$$\lambda_5 \text{ has ——— } 2.$$

$$\text{Clearly } \underline{n_1} + n_2 + \dots + n_k = n.$$

$$1 + 2 + 1 + 2 = 6.$$

$\text{null}(A - \lambda_{i_1} I)$ contributes n_1 linearly independent vectors.

$\text{null}(A - \lambda_{i_2} I)$ ——— n_2 ———

$\vdots$

$\text{null}(A - \lambda_{i_k} I)$ ——— n_k ———

Concatenate these lists (because vectors in diff eigenspaces above are going to be orthogonal if A is symmetric) to form a list of $n_1 + n_2 + \dots + n_k = n$ linearly indep vectors in $\mathbb{R}^n$.

Any vector in $\mathbb{R}^n$ is a linear combination of the above.

Let $v_1, \dots, v_n$ be n mutually orthogonal (unit-length) vectors.

$$A = V \Lambda V^T$$

$$V = \begin{bmatrix} v_1 & \dots & v_n \end{bmatrix}$$

$$\Lambda = \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix}$$

$$AV = V\Lambda.$$

$$V^T AV = (\underbrace{V^T V}_{=I})\Lambda = \Lambda = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}$$

* Interpret $A\underline{x}$ for any $\underline{x} \in \mathbb{R}^n$.

$$\underline{x} = \alpha_1 \underline{v}_1 + \dots + \alpha_n \underline{v}_n \quad (\text{because } \underline{v}_1, \dots, \underline{v}_n \text{ is a basis})$$

$$A\underline{x} = \alpha_1 A\underline{v}_1 + \alpha_2 A\underline{v}_2 + \dots + \alpha_n A\underline{v}_n.$$

$$= \alpha_1 \lambda_1 \underline{v}_1 + \alpha_2 \lambda_2 \underline{v}_2 + \dots + \alpha_n \lambda_n \underline{v}_n$$

(for symmetric matrices).

$$(A\underline{x})^T = \underline{x}^T A^T$$

* Suppose X : $m \times n$ matrix.

$$(AB)^T = B^T A^T.$$

$$(\underline{x}^T X)^T = \underline{x}^T (X^T)^T = X^T X. \quad (n \times n)$$

$$(X X^T)^T = (X^T)^T X^T = X X^T. \quad (m \times m)$$

$m \geq n$. (X is tall).

Eigenvalues of $X^T X$ are $\lambda_1, \dots, \lambda_n \geq 0$.

If $\lambda_n < 0$.

$$(X^T X) \underline{v}_n = \lambda_n \underline{v}_n.$$

$$\underline{v}_n^T (X^T X) \underline{v}_n = \lambda_n \underline{v}_n^T \underline{v}_n = \lambda_n \quad (\text{we chose } \underline{v}_n \text{ that way}).$$

$$\|\underline{w}_n\|^2 = \underline{w}_n^T \underline{w}_n = \underline{w}_n^T \underbrace{X \underline{v}_n}_{\underline{w}_n} = (\underline{v}_n^T X^T) (X \underline{v}_n).$$

Positive semi-definite.

* Eigenvalues of XX^T : Every e.v. of $X^T X$ is an e.v. of XX^T .

Let $\lambda_1, \dots, \lambda_n$ be e.v. of $X^T X$.

$$X X^T X \underline{v}_i = \lambda_i X \underline{v}_i.$$

$$\text{null}(X) = \text{null}(X^T X).$$

$$(X X^T) X \underline{v}_i = \lambda_i X \underline{v}_i.$$

$$\text{if } X^T X \underline{v}_i \neq 0.$$

$$\text{if } \underbrace{X^T X \underline{v}_i}_{\lambda_i \neq 0} \neq 0, \text{ then } X \underline{v}_i \neq 0.$$

means $\lambda_i > 0$. (because all e.v. of $x^T x$ are ≥ 0).

If $\lambda_i \neq 0$, then let $w_i = x v_i$ $w_i \neq 0$.

we have $x x^T w_i = \lambda_i w_i$

So w_i is an eigenvector of $x x^T$ with eigenvalue λ_i .

Suppose $\text{rank}(x) = k$. ($\leq \min(m, n)$).
 $\hookrightarrow$ # pivots in x .

free cols in x : $n - k$. $\therefore \dim(\text{null}(x)) = n - k$.

$\text{null}(x) = \text{null}(x^T x)$ $\therefore \dim(\text{null}(x^T x)) = n - k$.

multiplicity of eigenvalue 0 in $x^T x = n - k$.

$x x^T$: $\text{null}(x^T) = \text{null}(x^T x^T x^T) = \text{null}(x x^T)$.

x^T also has k pivots.

free cols in x^T : $m - k = (m - n) + n - k$.

$x x^T$ has $m - k$ free cols as well.

multiplicity of eigenvalue 0 in $x x^T = m - k = (m - n) + (n - k)$.

$\therefore x x^T$ and $x^T x$ share the same non-zero eigenvalues.

if $\lambda_i \neq 0$. v_i is an eigenvector of $x^T x$.

$$x^T x v_i = \lambda_i v_i$$

$$\therefore x x^T x v_i = \lambda_i x v_i \quad \text{Set } w_i = x v_i$$

$$x x^T w_i = \lambda_i w_i$$

If $\lambda_i \neq 0$ and w_i is an eigenvector of $x x^T$.

$$x x^T w_i = \lambda_i w_i$$

$$x^T x x^T w_i = \lambda_i x^T w_i \quad \text{Set } v_i = x^T w_i$$

$$x^T x v_i = \lambda_i v_i$$

Suppose u is a unit vector (vector with length 1)

Find u such that $\|x u\|$ is maximized.

For any vector $\|\underline{z}\|^2 = \underline{z}^T \underline{z}$

$$\|\underline{xu}\|^2 = (\underline{xu})^T (\underline{xu}) = \underline{u}^T \underline{x}^T \underline{x} \underline{u}.$$

Suppose eigen values of $\underline{x}^T \underline{x}$ are $\lambda_1 \geq \lambda_2 \dots \geq \lambda_n$.

Suppose $\underline{v}_1, \dots, \underline{v}_n$ are an orthonormal eigenvector basis for $\mathbb{R}^n$.

$$\underline{u} = \alpha_1 \underline{v}_1 + \dots + \alpha_n \underline{v}_n.$$

$$\underline{v}_i^T \underline{v}_j = \begin{cases} 0 & \text{if } i \neq j \\ 1 & \text{if } i=j. \end{cases}$$

$$\begin{aligned} \underline{x}^T \underline{x} \underline{u} &= \underline{x}^T \underline{x} (\alpha_1 \underline{v}_1 + \dots + \alpha_n \underline{v}_n) \\ &= \alpha_1 \underline{x}^T \underline{x} \underline{v}_1 + \dots + \alpha_n \underline{x}^T \underline{x} \underline{v}_n \\ &= \alpha_1 \lambda_1 \underline{v}_1 + \dots + \alpha_n \lambda_n \underline{v}_n. \end{aligned}$$

$$\begin{cases} A(2\underline{v}_1 + 3\underline{v}_2) \\ = A(2\underline{v}_1) + A(3\underline{v}_2) \\ = 2A\underline{v}_1 + 3A\underline{v}_2. \end{cases}$$

$$\begin{aligned} \underline{u}^T (\underline{x}^T \underline{x} \underline{u}) &= (\alpha_1 \underline{v}_1^T + \dots + \alpha_n \underline{v}_n^T) \cdot (\alpha_1 \lambda_1 \underline{v}_1 + \dots + \alpha_n \lambda_n \underline{v}_n) \\ &= \underbrace{\alpha_1^2 \lambda_1} + \alpha_2^2 \lambda_2 + \alpha_3^2 \lambda_3 + \dots + \alpha_n^2 \lambda_n. \end{aligned}$$

To maximise set $\alpha_1 = 1$. $\underline{u} = \underline{v}_1$.